SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – 26015**
- **Problem Statement Title – Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes**
- **Theme – Agriculture, FoodTech & Rural Development**
- **PS Category – Software**
- **Team ID – <your team ID>**
- **Team Name (Registered on portal) – <your team name>**

PRAMAAN ● प्रमाण

Turning geo-tagged watershed photographs into verifiable evidence.

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IDEA TITLE

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Proposed Solution (Describe your Idea/Solution/Prototype)

PRAMAAN — an evidence-reconciliation layer over DoLR's existing SRISHTI / DRISHTI geotag pipeline. Every field photograph becomes a claim the satellite record can test.

CLAIM › EVIDENCE › VERDICT › OFFICER DECIDES › SIGNED RECORD

PHOTO 0.15 | TERRAIN 0.25 | SATELLITE 30 m 0.25 | TEMPORAL + CONTROLS 0.25 | RAINFALL CONTEXT 0.10

Detailed explanation of the proposed solution

- Each DRISHTI geotag — lat/lon, GPS accuracy, orientation, timestamp, activity, status — becomes a structured, testable claim
- Reconciled against 5 independent evidence families: photo AI, terrain from DEM, satellite indices at 30 m, temporal trend, rainfall context
- Output: a verdict on a published L0–L4 evidence ladder, with calibrated confidence and a mandatory dissent panel
- Officer accepts / edits / rejects; signed into an append-only, hash-chained ledger
- Deliverable: an Evidence Pack PDF carrying every scene ID, model version and weight used

How it addresses the problem

- Today a geotag is moderated by eye — accept or reject. That verifies a photograph, not an outcome
- 1,24,830 water-harvesting structures under WDC-PMKSY 2.0 cannot be eyeballed at scale
- Implements the WDC-PMKSY 2.0 written mandate: NDVI/NDWI change detection, and cross-verification of satellite images with ground interventions
- Converts ~1,200 works per district into ~40 that need a human decision, each with a documented reason
- Adds one step to the existing workflow — replaces no government system

Innovation and uniqueness of the solution

- Matched-control differencing per structure — 5–12 similar un-treated sites in the same sub-watershed, so we measure the intervention, not the monsoon
- Terrain plausibility screen — a check dam claimed off any drainage line is flagged deterministically, before any AI runs
- Detectability gate — the system refuses to judge structures smaller than one 30 m pixel and escalates to a cluster claim
- Causal claims are unreachable in code; "requires physical verification" is the strongest language it can emit
- Every officer correction becomes labelled training data

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TECHNICAL APPROACH

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Technologies to be used (e.g. programming languages, frameworks, hardware)

BACKEND

Python 3.11 · FastAPI · Celery · Redis · Docker Compose

GEOSPATIAL

PostgreSQL + PostGIS · GDAL · rasterio · GeoPandas · Shapely · PyProj · WhiteboxTools

AI / CV

PyTorch · SigLIP-2 zero-shot (no labelled corpus exists) · OpenCV · Prithvi-EO-2.0 30 m foundation model as the research path

FRONTEND

React · TypeScript · MapLibre GL · TiTiler for COG tiles

DATA SOURCES

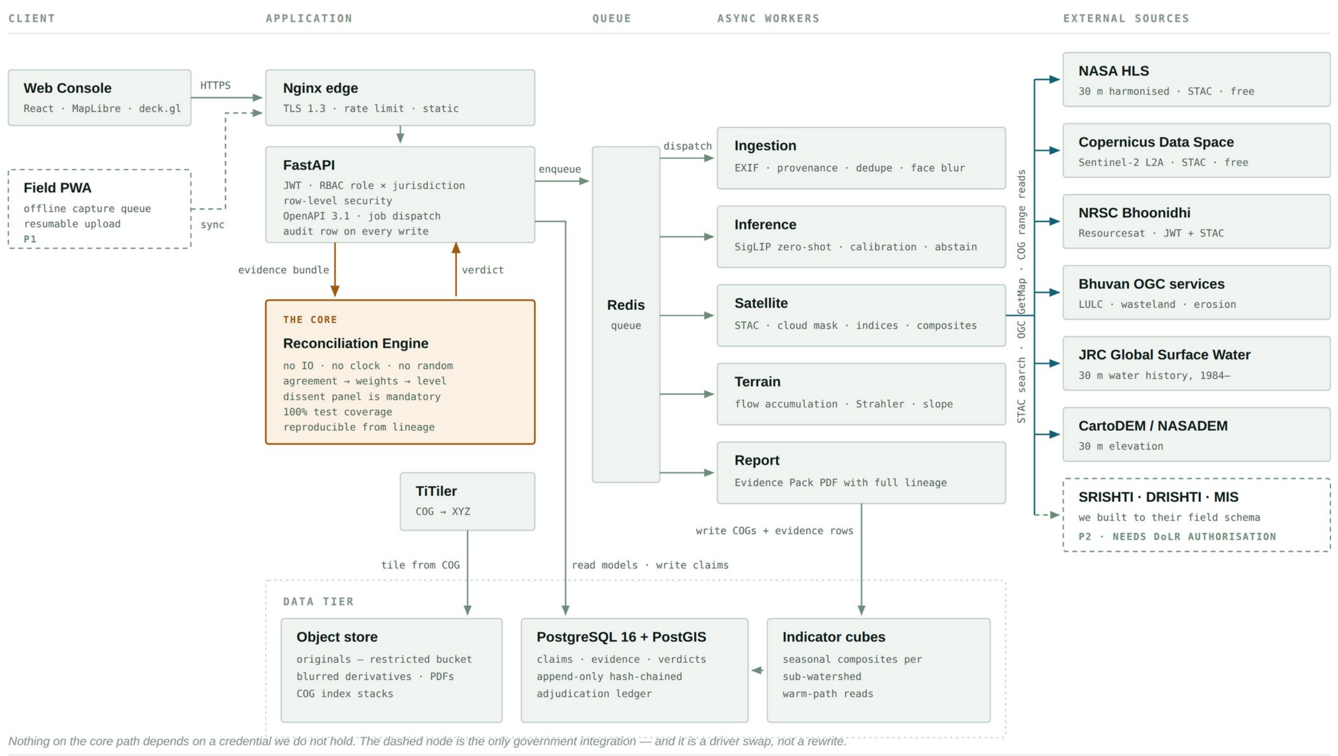
NASA HLS 30 m (STAC) · Copernicus Sentinel-2 (STAC) · NRSC Bhoonidhi API (JWT + STAC) · Bhuvan OGC WMS · JRC Global Surface Water · CartoDEM · SLUSI watershed atlas

HARDWARE

One 8-core VM. No GPU required on the core path. ₹0 in data and licence cost.

Methodology and process for implementation (Flow Charts / Images / working prototype)

INGEST › VALIDATE › INTERPRET › RECONCILE › ADJUDICATE › REPORT



Runtime architecture. The reconciliation engine is a pure function — an evidence bundle in, a verdict out — so any verdict is reproducible from its stored lineage.

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

- Every core data source is free, open and programmatic — endpoints verified, not assumed
- NRSC Bhoonidhi exposes a documented JWT + STAC API; data coarser than 5 m is open under Indian Space Policy 2023
- ₹0 in data and licence cost; entire stack is open-source
- Runs on a single 8-core VM with no GPU on the core path
- Built to NRSC's published DRISHTI field schema, so production integration is a driver swap, not a rewrite
- Complete workflow demonstrable offline from a local cache

Potential challenges and risks

- Many structures are smaller than one 30 m pixel — farm ponds, gully plugs, individual trenches
- Monsoon cloud cover can destroy an entire kharif observation window
- EXIF GPS is often stripped by messaging apps; coordinates may be typed manually
- Seasonality and rainfall can be mistaken for genuine intervention impact
- A false negative could unfairly implicate a beneficiary
- No public SRISHTI / DRISHTI API could be found during research

Strategies for overcoming these challenges

- Detectability gate compares footprint to sensor resolution first, then escalates to a cluster claim — never a false negative
- Rabi and summer windows carry the analysis; cloud is scored per area of interest, not per scene
- Ranked provenance chain (EXIF → sidecar → CSV → manual pin) plus a deterministic terrain plausibility screen
- Matched controls in the same sub-watershed, same-season comparison, construction window excluded
- Verdicts stay provisional until a named officer signs; wording is locked to "requires physical verification"
- Imagery and geotag sources sit behind driver interfaces — one class to swap

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IMPACT AND BENEFITS

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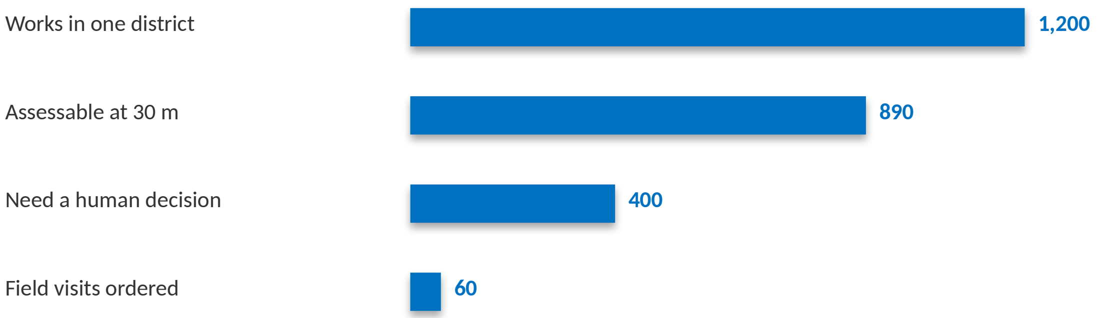
Potential impact on the target audience

- WCDC Project Manager — ~1,200 works per district reduce to ~40 needing a decision each month, each with a documented reason for the field visit
- SLNA monitoring & evaluation — a standing, reproducible evidence pipeline replaces hand-built one-off GIS analyses
- DoLR — every aggregate number traces to a signed adjudication, making progress reporting audit-defensible
- Field WDT — fewer wasted verification journeys; feedback on capture quality

TARGETING SCARCE VERIFICATION EFFORT — ONE DISTRICT, ONE MONTH



Benefits of the solution (social, economic, environmental, etc.)

ECONOMIC

- Targets a scarce verification budget inside a ₹8,134 crore programme covering 49.5 lakh hectares
- ₹0 data and licence cost — marginal cost per extra structure is CPU seconds

SOCIAL

- Accountability without accusation — the system never says fraud or false
- Strongest wording it can emit is “requires physical verification”; a named officer makes the finding

ENVIRONMENTAL

- Earlier detection of non-performing soil and water conservation works
- Faster remedial action on degraded land, water retention and vegetation cover

INSTITUTIONAL

- An expert-labelled dataset of Indian watershed field evidence grows from routine use
- Owned by the department, at no additional cost

Watershed development is already proven: mean benefit–cost ≈ 2, IRR 27.4%, cropping intensity +35.5% (ICRISAT, 636 Indian micro-watersheds). The gap is knowing which structures work.

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RESEARCH AND REFERENCES

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Details / Links of the reference and research work

GOVERNMENT OF INDIA — PROGRAMME & PLATFORM

- WDC-PMKSY 2.0 portal and national dashboard — wdcpmksy.dolr.gov.in
- Guidelines for New Generation Watershed Development Projects (WDC-PMKSY 2.0), DoLR — mandates NDVI/NDWI change detection and cross-verification of satellite images with ground interventions
- National Technical Guidelines for Improved Watershed Management, NRAA, Aug 2025
- NRSC SRISHTI–DRISHTI User Manual and DRISHTI v2.3 Manual — bhuvan-app1.nrsc.gov.in/iwmp
- Bhuvan Watershed Development Component 2.0 — bhuvan-app1.nrsc.gov.in/wdc2.0
- MoRD press release, DoLR–NRSC MoU, 18 Oct 2023 — SRISHTI 2.0 and DRISHTI 2.0
- PIB: one crore MGNREGA assets geotagged (PRID 1488368)

DATA SOURCES & SCIENTIFIC LITERATURE

- Bhoonidhi API Specification, NRSC — bhoonidhi.nrsc.gov.in/bhoonidhi-api (JWT + STAC)
- NASA Harmonized Landsat Sentinel-2 (HLS) 30 m v2.0 — LP DAAC
- Copernicus Data Space Ecosystem STAC catalogue
- JRC Global Surface Water v1.4 (Pekel et al.) — 30 m water history, 1984–
- SLUSI Digital Watershed Atlas of India
- Prithvi-EO-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation — arXiv:2412.02732 (pretrained on 4.2 M HLS samples at 30 m)
- Assessing impacts of watershed interventions using ground data and remote sensing, Ethiopia — Springer, doi 10.1007/s13762-021-03192-7
- Impact of Watershed Program and Conditions for Success: A Meta-Analysis — ICRISAT, 636 micro-watersheds
- Small water bodies mapped from Sentinel-2 MSI — Int. J. Remote Sensing 41(20)